

Q1a).

```
?crabs
```

The `crabs` data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

Source: Campbell, N.A. and Mahon, R.J. (1974) A multivariate study of variation in two species of rock crab of genus *Leptograpsus*. *Australian Journal of Zoology* **22**, 417–425.

Q1b). Loading the data and displaying its structure.

Q1c) to produce frequency table

```
Frequency<-table(crabs$sex)
Frequency
F      M
100 100
```

D) to construct bar plot

```
barplot(Frequency,main = "Morphological Measurement of Crabs",
        xlab = "Sex",
        ylab = "Frequency", col=c(1:2))
```

From Figure 1, the barplot, the sex appear to have approximately equal number of observations. This suggest that the dataset is balanced across the two sexes in the crabs dataset whichh contains 200 observations. One species does not outweigh the others in frequency.

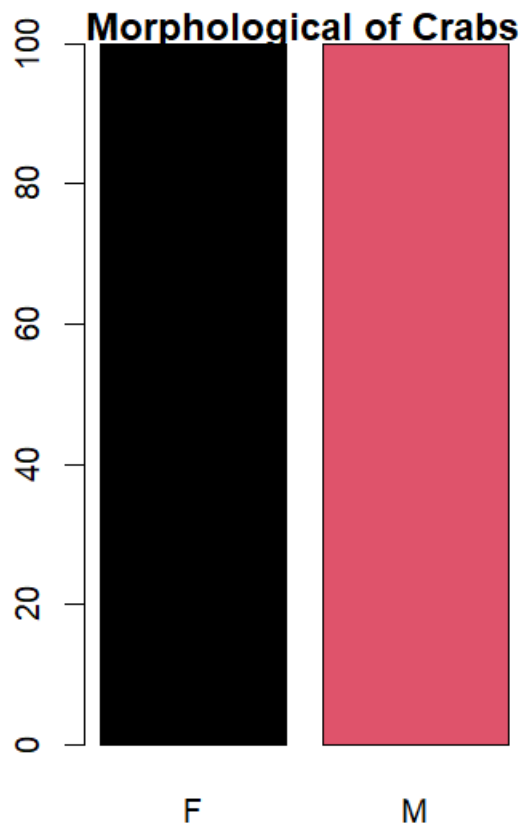
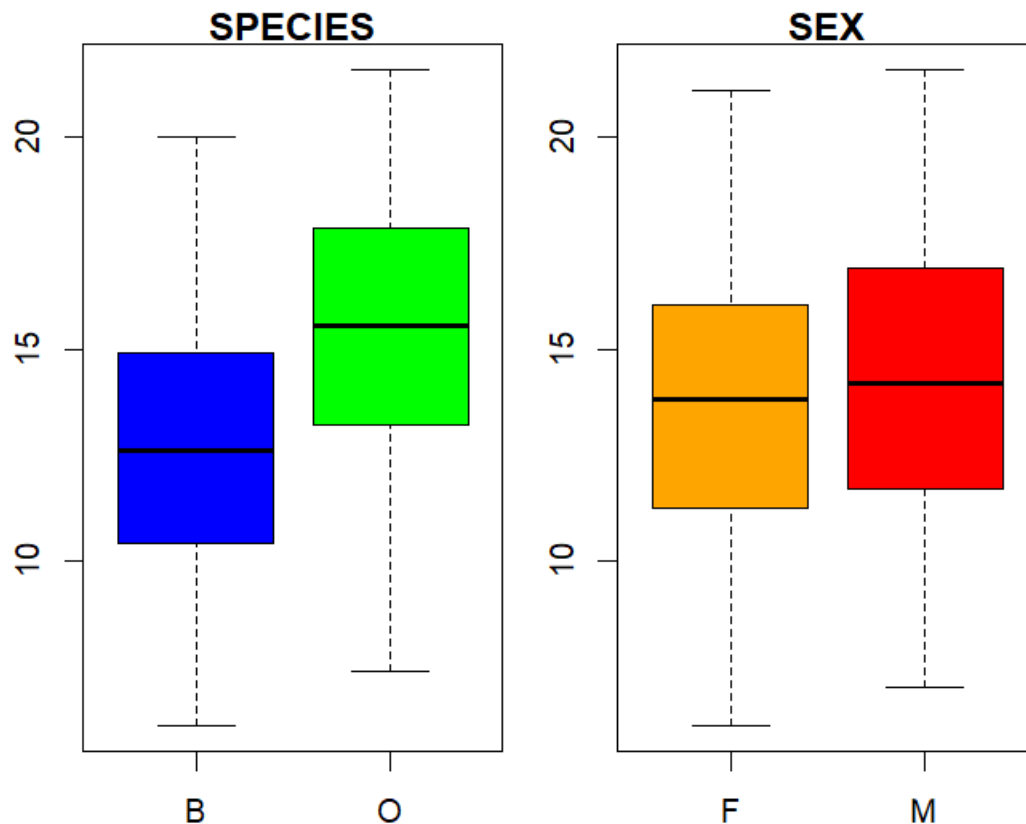


Figure 1: Morphology of crab

Q2)

```
par(mfrow=c(1,2))  
#to construct a boxplot of body depth for the four groups  
boxplot(crabs$BD~crabs$sp, data=crabs, main="SPECIES",  
        col=c("blue","green"))  
  
boxplot(crabs$BD~crabs$sex, data=crabs, main="SEX",  
        col=c("orange","red"))
```



B) outliers: Male species typically shows few small outliers in body depth.

Female species generally shows no extreme outliers in this variable

Comparison: All the species have the same mean and median which show that they distribution is symmetric

```
mean_O<-mean(crabs$BD[crabs$sp=="O"])
mean_B<-mean(crabs$BD[crabs$sp=="B"])
median_O <- median(crabs$BD[crabs$sp=="O"])
median_B <-median(crabs$BD[crabs$sp=="B"])

mode_O<- crabs$BD[which.max(table(crabs$BD[crabs$sp == "O"]))]
mode_B<- crabs$BD[which.max(table(crabs$BD[crabs$sp == "B"]))]
var_O<-var(crabs$BD[crabs$sp=="O"])
var_B<-var(crabs$BD[crabs$sp=="B"])

Central_values<-data.frame(
  mean=c(mean_O,mean_B),
```

```

median=c(median_O, mode_B),
mode=c(mode_O,mode_B),
var=c(var_O,var_B)
)
print(Central_values)

```

Result

	mean	median	mode	var
1	15.478	15.55	7.7	9.931834
2	12.583	8.20	8.2	9.411930

Q3)

```

par(mfrow=c(1,2))

hist(crabs$CL[crabs$sp== "O"],main = "SPECIES O", xlab="lenght
(cm)",
      breaks = 3,
      border = "white",col="darkgreen")

hist(crabs$CL[crabs$sp== "B"],main = "SPECIES B", xlab="lenght
(cm)",
      breaks = 3,
      border = "white",col="darkred")

```

From figure 2, Skewness: species O is negatively skewed, species B is symmetric since it has a bell shape

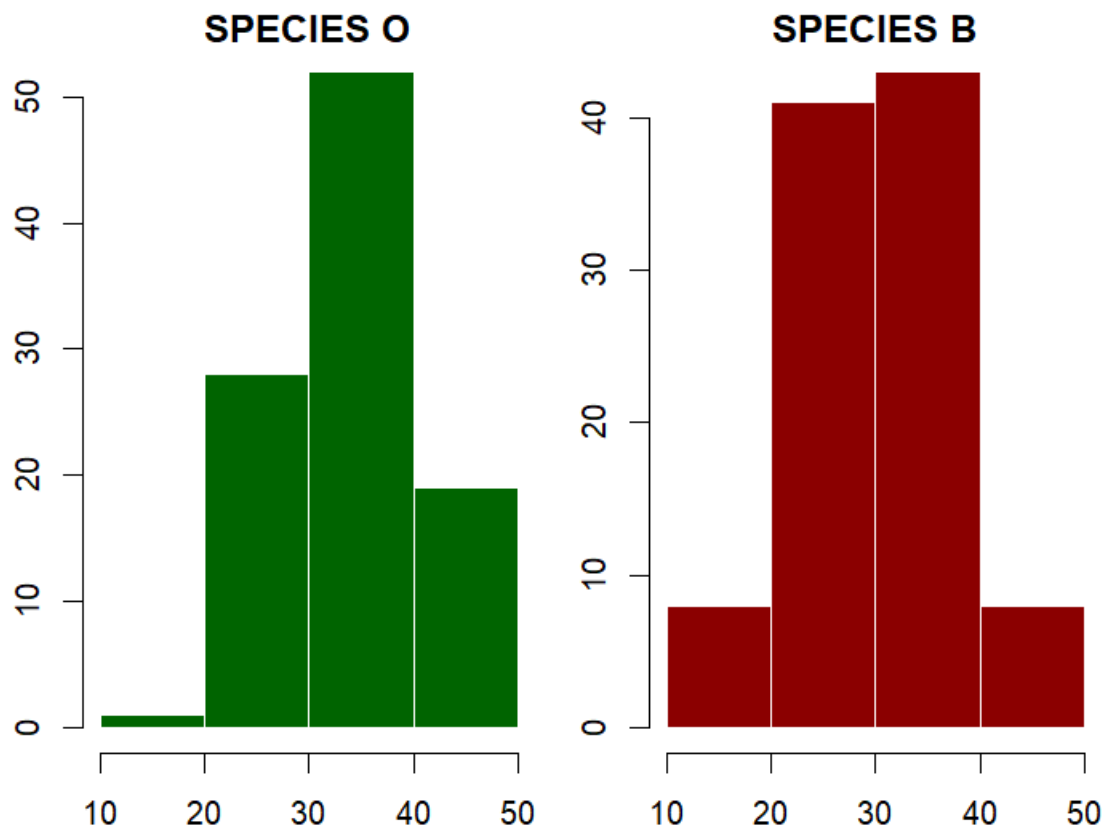
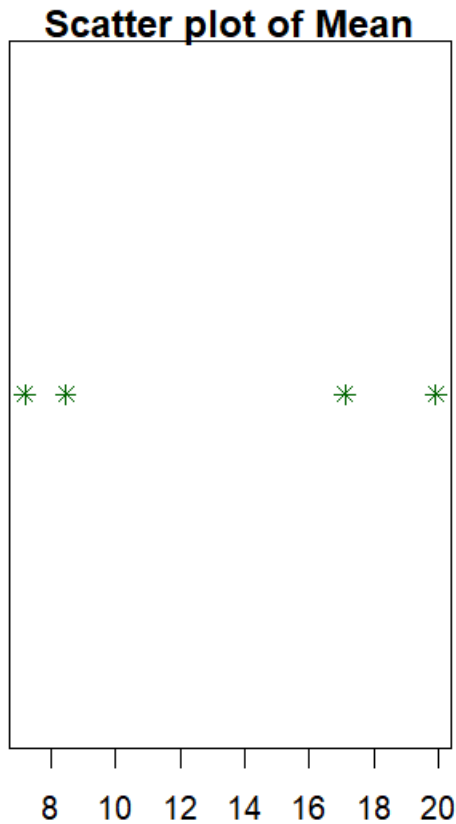


Figure 2: Bar plot

Q4)

	mean
1	7.20
2	19.90
3	7.20
4	17.10
5	8.45

Q5A



Q6)

Introduction

The crabs dataset, introduced by Campbell, N.A. and Mahon, R.J. in the year 1974, contains morphological measurements of two crabs species: "O", and "B."

The aim of this report is to conduct a descriptive statistical analysis of the dataset using R and to examine differences among species.

Analysis

The dataset contains 200 observations evenly distributed across two species, making it balanced for comparative analysis.

Measures of central tendency(mean) indicate that

morphological measurements does not vary substantially across species,

Conclusion

The crabs dataset does not demonstrates clear morphological separation among the two species.